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1) Introduction to Cryptographic Systems

Security refers to the combination of tools and technologies used to defend an organization's digital assets. The primary objective of security is to protect the system from unauthorized access, misuse, disclosure, modification, or destruction. In today's world security is really very important as it ensures employees safety, ensures business continuity, protects sensitive information, deters crime and unauthorized access (software, 2025). With the rapid expansion of digital communication and cloud-based storage, sensitive information is constantly exposed to potential cyber threats. Therefore, security mechanisms are essential to safeguard information throughout its lifecycle, from creation and storage to transmission and disposal.

According to (Stallings, 2017), information security is concerned not only with preventing unauthorized access but also with ensuring that information remains trustworthy and accessible to legitimate users. A lack of proper security controls can result in data breaches, financial losses, and damage to organizational reputation.

1.1) CIA triad

The CIA triad is a fundamental concept in cybersecurity that defines the three main objectives of information security: confidentiality, integrity, and availability. These principles work together to ensure that information system remains secure, reliable and trustworthy. By applying the CIA triad, organizations can protect sensitive data from unauthorized access, prevent unwanted modifications, and ensure that information is accessible to authorized users whenever required.

To support the CIA triad, security mechanisms such as authentication, authorization and accounting are commonly used. These mechanisms help verify user identities, control access rights, and monitor system activities. Together, they play a vital role in maintaining effective information security.

- **Confidentiality:** Confidentiality focuses on ensuring that sensitive data is accessible only to permitted users. Cryptographic methods, particularly encryption,

protect confidentiality by converting readable information into an encoded form that cannot be understood without the appropriate decryption key. (Schneier, 1996)

- **Integrity:** Integrity refers to preserving the accuracy, reliability, and completeness of information throughout its lifecycle. Data must remain unchanged during storage or transmission, and safeguards should be in place to prevent unauthorized modification (Hashemi-Pour, 2023).
- **Availability:** Availability ensures that authorized users can access information and systems whenever required. This requires dependable hardware, secure network infrastructure, and well-maintained systems that support continuous access to data (Hashemi-Pour, 2023).

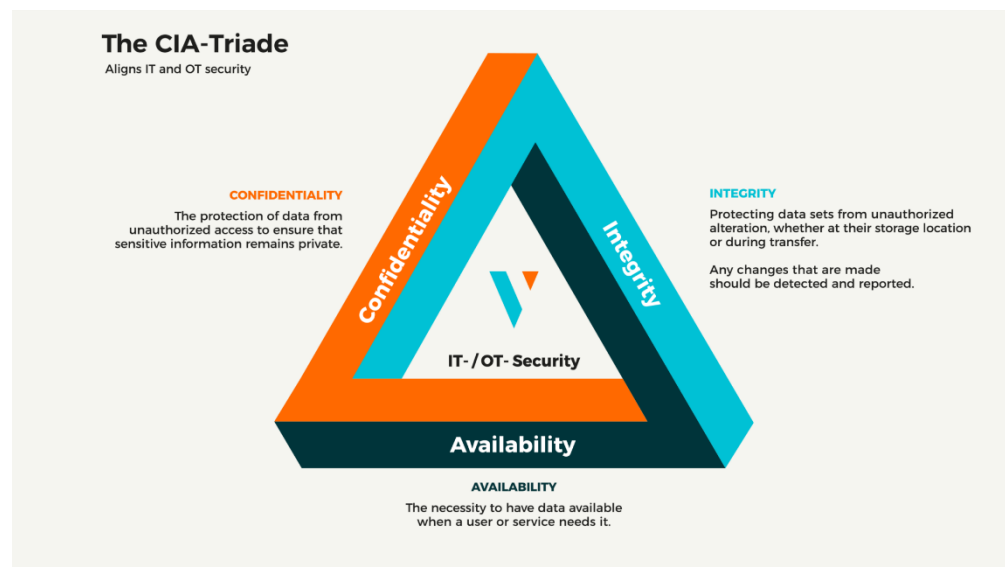


Figure 1: CIA Triad (Redder, 2025)

1.2) Cryptography

The term cryptography originates from the Greek words **kryptos**, which means hidden, and **graphein**, which means writing. Together, these words describe the concept of concealed or secret writing. Cryptography refers to the practice of securing messages or communications between individuals or groups so that the information remains confidential and inaccessible to unauthorized parties.

The need for cryptography arose during early human civilization when people began forming tribes and organized societies. As competition for power and control increased among different groups, secure communication became essential. This growing demand for secrecy in communication led to the development of early cryptographic techniques, which laid the foundation for modern cryptography. (Journals, n.d.).

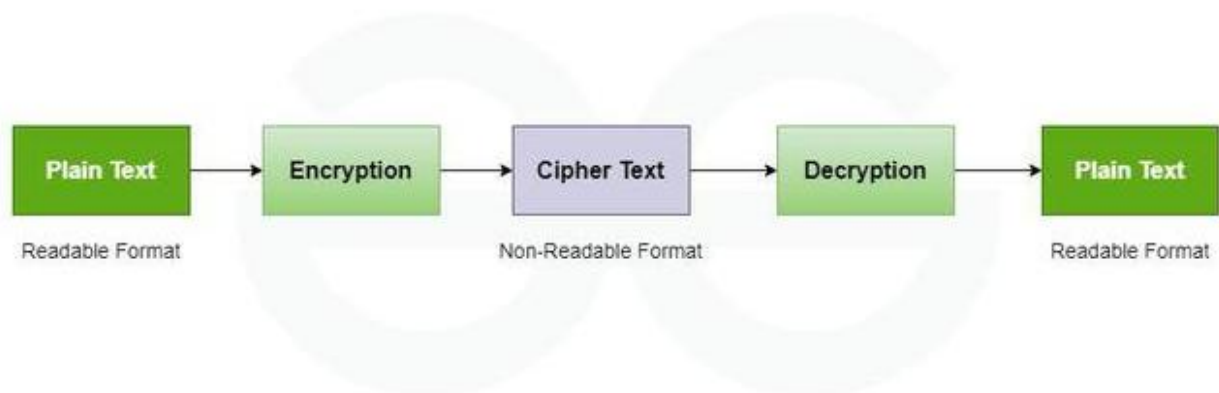


Figure 2 :Basic Cryptography Communication Model (GeeksforGeeks, 2025)

1.3) Key terminologies of cryptography

Table 1: Table of key terminologies

Keys	Definition
Public key	A cryptographic key that is freely distributed and used in public-key (asymmetric) encryption systems to secure data.
Private key	A secure and restricted key in asymmetric cryptography, kept secret by the owner and used to decrypt information or create digital signatures.
Encryption	The technique of transforming readable data into an encoded format using a cryptographic key to prevent unauthorized access.
Decryption	The procedure of restoring encrypted data back to its original readable form by applying the correct key.
Cipher Text	The scrambled and unintelligible output produced after data has been encrypted.
Plain Text	The original, human-readable data before encryption or the recovered data after decryption.
Secret Key	A single shared key utilized in symmetric encryption, where the same key is used for both encryption and decryption. (Mishra, et al., 2021)

1.4) History of Cryptography

1.4.1) Ancient Origins of Cryptography

The term cryptography comes from the ancient Greek words **kryptos**, meaning “hidden,” and **graphein**, meaning “writing.” Together, they describe the practice of secret writing used to protect information from unauthorized readers. Long before the invention of computers, early civilizations recognized the need for secure communication, especially in areas such as leadership, diplomacy, and warfare (Naser, 2021).

Some of the earliest known examples of cryptography date back to around 1900 BCE in ancient Egypt, where secret inscriptions were carved into stone. One well-known early method is the Spartan scytale, a transposition technique that required wrapping a strip of parchment around a staff of a specific size to reveal the hidden message. Similarly, Julius Caesar later introduced a substitution technique now known as the *Caesar cipher* by shifting letters of the alphabet to encode military messages (Naser, 2021).



Figure 3: Ancient Cryptography

1.4.2) Medieval Development of Cryptography

As societies expanded and literacy increased, cryptographic techniques became more structured and widespread. Many early ciphers across different cultures relied on either substituting letters or rearranging their order. Comparable methods were observed in Hebrew and Indian cryptographic traditions, where letters were altered based on position or pairing to obscure meaning (Naser, 2021).

A major advancement occurred during the medieval period in the Islamic world. In the 9th century, the scholar Al-Kindi introduced *frequency analysis*, a mathematical technique used to break substitution ciphers. This marked the beginning of cryptanalysis as a scientific discipline and significantly influenced the future development of cryptographic methods (Naser, 2021).

Papa	aut.	Vallis sporgians	sunt	Tureisclara	hanc
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Rex Aragonu	plus	Niuatrum	M	Thizarnum	id
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Placenna	velut	Officiales	tot		
Laude	pp	Officialis	...		
Ilm. p. Blanca Maria	genf	Dolrad Torcius	vil		
Filinum	gg	Co. Jacob Torcius	sub		
Sanchus secundus	qz	Xpoforus Tanelus	ab		
Cornilium	sup	Corniliaro	dant		
Dofaus	ep	Petra teeta	cor		
Decorum	Inde	N. d. Joha d. Rubois	quot		
Basilia nola	ao	Pereus mara	victor		
Ripalta	q	Co. Aneto d. S. virali	trans		
Capreignam	qdem	Co. Scipio d. S. virali	quon		
Pignotulum	zif	Fontanollate	X		
Beducium	stat	Nucetum	X		
Baredonum	fecr	Tecentium	X		

Figure 4: Medieval Development of Cryptography (Cipher Mysteries, 2016)

1.4.3) Modern Evolution and Applications of Cryptography

During the Renaissance and early modern era, cryptography continued to advance with the introduction of more complex systems, such as polyalphabetic ciphers like the

Vigenère cipher. Mechanical cipher devices were also developed to automate encryption processes. In the 20th century, cryptography played a critical role in warfare, with machines such as the German Enigma being used to secure military communications (Naser, 2021).

In the modern digital age, cryptography has shifted toward mathematically driven and algorithm-based systems. Contemporary cryptography relies on fields such as number theory, computational complexity, and information theory. It now supports essential technologies including public-key cryptography, digital signatures, secure communication protocols, and data protection systems that are fundamental to cybersecurity today (Naser, 2021).

1.5) Encryption

According to the National Institute of Standards and Technology (NIST), encryption is a cryptographic technique that safeguards information by transforming human-readable data, referred to as plaintext, into an encoded form known as ciphertext. This process conceals the original content of the data from unauthorized parties and allows access only to individuals who hold the appropriate cryptographic key. Encryption is essential for preserving data confidentiality, particularly when information is stored or transmitted across unsecured networks. (Technology, n.d.)

1.5.1) Symmetric Encryption

Symmetric encryption is a cryptographic technique in which a single shared secret key is used to both secure and recover information. The sender encrypts the readable data, known as plaintext, into an encoded form called ciphertext using this key. The recipient then applies the same key to decrypt the ciphertext back into its original readable state. This type of encryption is highly efficient and performs quickly, making it particularly effective for protecting large volumes of data. However, a significant weakness of symmetric encryption is the challenge of safely distributing the secret key, as unauthorized access to the key can lead to a complete loss of communication security. Widely used symmetric encryption algorithms include the Advanced Encryption Standard (AES) and the Data Encryption Standard (DES). (J.Simmons, 1979).

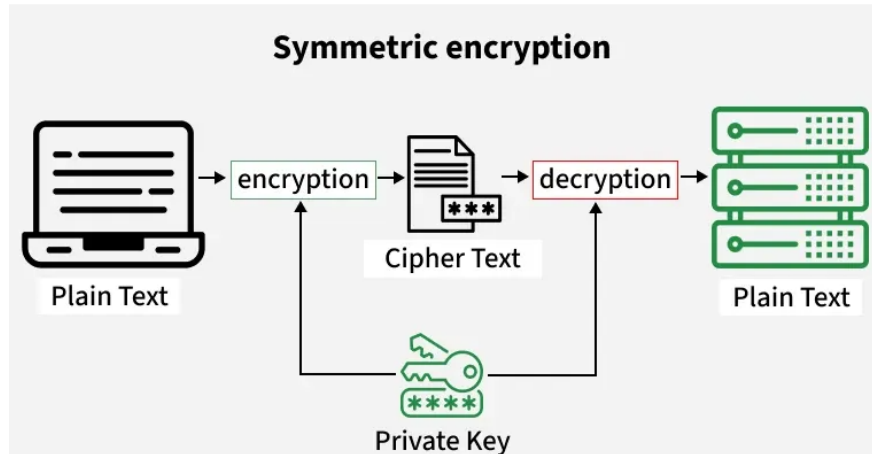


Figure 5: Symmetric Encryption (GeeksforGeeks, 2025)

1.5.2) Asymmetric Encryption

Asymmetric encryption, commonly referred to as public-key cryptography, operates using two mathematically linked keys: a public key and a private key. The public key is used to encrypt data and can be freely shared, while the private key is kept confidential and is required to decrypt the encrypted information. Unlike symmetric encryption, this approach removes the need for exchanging a secret key in advance. In addition to ensuring confidentiality, asymmetric encryption supports authentication and digital signatures, which help verify the identity of the sender and maintain message integrity. Although this method is computationally slower than symmetric encryption, it is typically used to securely transfer a shared secret key, which is then employed for efficient symmetric encryption of the actual data. Common asymmetric encryption algorithms include RSA and Elliptic Curve Cryptography (ECC). (J.Simmons, 1979).

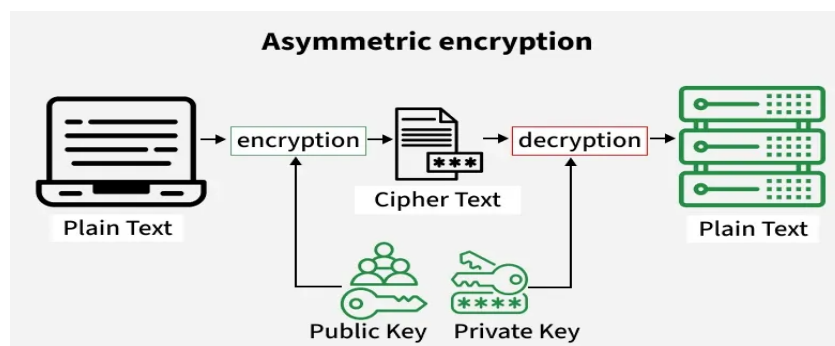


Figure 6: Asymmetric encryption (GeeksforGeeks, 2025)

1.6) Decryption

Decryption is the reverse process of encryption, where encoded or encrypted data (ciphertext) is transformed back into its original readable form (plaintext) using a specific key. It ensures that only authorized users who possess the correct key can access the original information, maintaining confidentiality and data integrity. Decryption is an essential component of modern cryptography, enabling secure communication in digital systems. (Technology, n.d.)

1.6.1) Symmetric Decryption

Symmetric decryption is the process of converting ciphertext back into plaintext using the same key that was used for encryption. In symmetric cryptography, both the sender and receiver share a single secret key, which must remain confidential. Symmetric decryption is fast and efficient, making it ideal for encrypting large amounts of data, but it requires secure key distribution between parties. (Stallings, 2017)

1.6.2) Asymmetric Decryption

Asymmetric decryption is the process of converting ciphertext into plaintext using a different key from the one used for encryption. In asymmetric cryptography, two keys are used: a public key for encryption and a private key for decryption. Only the recipient who possesses the private key can decrypt the message, ensuring secure communication even if the public key is widely shared. This method is computationally heavier than symmetric decryption but provides enhanced security for key exchange. (Stallings, 2017)

2) General of Caesar Cipher

The Caesar Cipher is among the earliest and most straightforward techniques used for encrypting information. It is named after the Roman leader Julius Caesar, who is believed to have used this method to keep his military messages confidential. The cipher works by replacing each letter in a message with another letter that is a fixed number of positions ahead in the alphabet. For instance, when a shift value of three is applied, the letter **B** is replaced with **E**. (GeeksforGeeks, 2025).

2.1) Background of Caesar Cipher

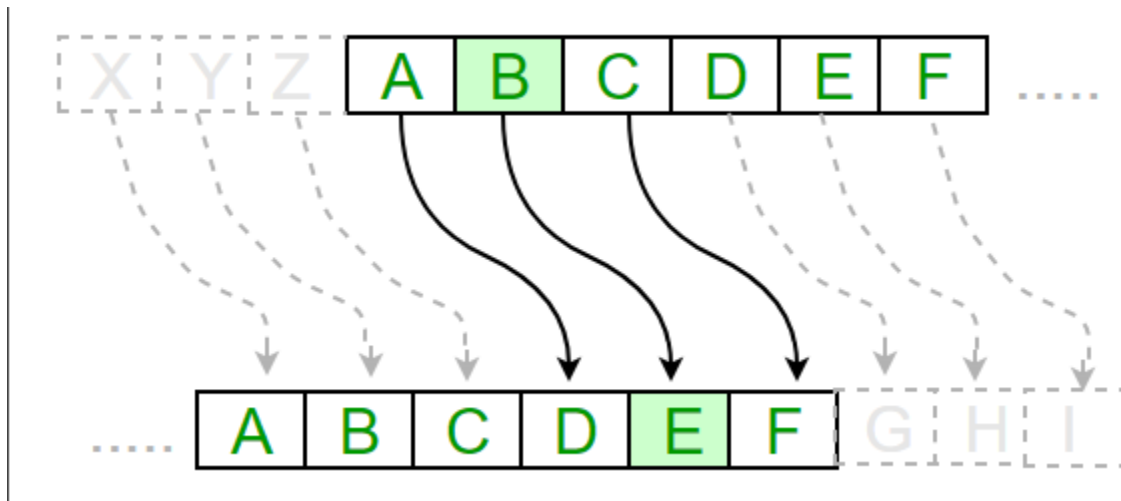


Figure 7: Caesar Cipher (GeeksforGeeks, 2025)

The Caesar Cipher is one of the earliest known cryptographic techniques and is widely recognized as a foundational method in the history of secure communication. The cipher is named after Julius Caesar, who is documented to have used this method to protect military correspondence during his campaigns, ensuring that sensitive messages could not be understood if intercepted by enemies (Britannica, 2024). This highlights the early use of cryptography to achieve confidentiality in warfare.

The Caesar Cipher is categorized as monoalphabetic substitution cipher and operates under symmetric key cryptography, meaning the same key is used for both encryption and decryption (Babylon, n.d.). The technique functions by shifting each letter of the plaintext forward by a fixed number of positions in the alphabet. This fixed value, known as the shift key, determines the mapping between plaintext and ciphertext characters. When the alphabet reaches its end, the shifting process wraps around to the beginning, maintaining a continuous substitution pattern (Babylon, n.d.).

Due to its simplicity, the Caesar Cipher was highly practical in ancient times, as it required minimal effort and no specialized tools to perform encryption and decryption. According to (Silva-Santos, 2024) this simplicity allowed rapid message encoding, which was particularly beneficial in military environments where timely communication was critical. However, the limited number of possible shift values results in a small key space, making

the cipher vulnerable to brute-force attacks and basics cryptanalysis techniques such as frequency analysis (Verma & kaur, 2014).

Despite this weakness, the Caesar Cipher holds significant historical importance in the evolution of cryptographic systems. It introduced the core idea of substitution-based encryption, which later influenced the development of more advanced classical ciphers and modern cryptographic thinking (ENSAE, n.d.). Today the Caesar Cipher remains an important educational tool in cybersecurity and cryptography studies, as it provides a clear and accessible introduction to key-based encryption concepts and the fundamental goals of secure communication (Britannica, 2024).

2.2) Encryption and Decryption process of Caesar Cipher

2.2.1) Encryption process of Caesar cipher

The encryption process in the Caesar Cipher converts plain text into cipher text by shifting each letter of the message forward by a fixed number of positions in the alphabet. First, a shift value (also called a key) is chosen, such as 3. Then, each letter in the original message is replaced with another letter that appears three places ahead in the alphabet. If the shift moves beyond the letter Z, it wraps around to the beginning of the alphabet. This process continues until all letters in the message are transformed, resulting in an encrypted message that hides the original content from unauthorized users.

Logical Explanation of Encryption Process

The encryption logic of the Caesar Cipher works by shifting each letter of the plaintext forward by a fixed number of positions in the alphabet to produce the ciphertext.

Mathematical Explanation (Formulae) of Encryption Process

The encryption formula is:

$$C = (P + K) \bmod 26$$

Where,

- P = numerical value of the plaintext letter
- K = shift value (key)
- C = numerical value of the ciphertext letter

2.2.2) Decryption process of Caesar Cipher

The decryption process is the reverse of encryption and is used to recover the original message from the cipher text. In this step, the same shift value used during encryption is applied in the fixed number of positions in the alphabet. If the shifting goes before the letter A, it wraps around to the end of the alphabet. By applying this backward shift to every letter, the original plain text message is successfully restored.

2.2.2.1) Logical Explanation of Decryption Process

The decryption logic reverses the encryption process by shifting each letter of the ciphertext backward by the same fixed number of positions to recover the original message.

2.2.2.2) Mathematical Explanation (Formulae) of Decryption Process

The Decryption formula is:

$$P = (C - K) \bmod 26$$

Where,

- C = numerical value of the ciphertext letter
- K = shift value (key)
- P = numerical value of the plaintext letter

2.2.3) Example of Encryption and Decryption using Caesar Cipher

- Plaintext: COOKIE
- Key: 3

For Encryption,

$$C = (P + K) \bmod 26$$

Each letter of the plaintext is shifted forward by three positions in the alphabet to produce the ciphertext.

Table 2: Encryption process of traditional Caesar Cipher

Plaintext	Alphabet Position	Shifted by +3	Ciphertext Letter
C	2	5	F
O	14	17	R
O	14	17	R
K	10	13	N
I	8	11	L
E	4	7	H

Ciphertext: FRRNLH

After applying the Caesar Cipher encryption with a shift of 3, the plaintext **COOKIE** is encrypted as **FRRNLH**.

For Decryption,

$$P = (C - K) \bmod 26$$

To recover the original message, the decryption process shifts each letter of the ciphertext backward by the same key value used during encryption.

- **Ciphertext: FRRNLH**

Plaintext	Alphabet Position	Shifted by +3	Ciphertext Letter
F	5	2	C
R	17	14	O
R	17	14	O
N	13	10	K
L	11	8	I
H	7	4	E

Table 3 : Decryption Process of Traditional Caesar Cipher

Recovered Plaintext: COOKIE

This confirms that the original plaintext is successfully restored using the same shift key, demonstrating the correct working of both encryption and decryption processes in the Caesar Cipher.

2.3) Advantage of Caesar Cipher

The Caesar Cipher has many advantages, some of them are:

- **Simple and easy to understand:** The Caesar Cipher uses a straightforward letter-shifting technique, making it easy to learn and implement, especially for beginners in cryptography (Babylon, n.d.)
- **Fast encryption and decryption:** Since the algorithm involves only basic arithmetic shifting, encryption and decryption operations can be performed very quickly (Mishra, et al., 2021)
- **Low computational and resource requirement:** The cipher does not require complex calculations or advanced computing resources, allowing it to be executed manually or with minimal tools (Verma & kaur, 2014)
- **Foundation for cryptographic learning:** The Caesar Cipher introduces fundamental cryptographic concepts such as keys, substitution, encryption, and decryption, making it an important teaching tool in cybersecurity education (Britannica, 2024)
- **Historical Importance:** As one of the earliest encryption methods, the Caesar Cipher played a significant role in the evolution of cryptography and influenced the development of more advanced classical ciphers (ENSAE, n.d.)

2.4) Disadvantages of Caesar Cipher

Despite its simplicity, the Caesar Cipher has several weaknesses that limit its effectiveness in modern security systems.

- **Very small key space:** The cipher allows only 25 possible shift values, making it highly vulnerable to brute-force attacks (Mishra, et al., 2021)
- **Susceptible to frequency analysis:** Because the same substitution pattern is used throughout the message, attackers can easily analyse letter frequencies to break the cipher (Verma & kaur, 2014)
- **Lack of Cryptographic Strength:** The predictable structure of the Caesar Cipher provides no resistance against modern cryptographic attacks (Babylon, n.d.)
- **No protection against modern threats:** The cipher does not meet current security requirements and is unsuitable for protecting digital communications (Britannica, 2024)
- **Not suitable for sensitive or confidential data:** Due to its weak security design, the Caesar Cipher should not be used for encrypting important or confidential information (ENSAE, n.d.)

3) Development of a new Cipher

3.1) Need for Modifying the Traditional Caesar Cipher

The Caesar cipher is one of the earliest and simplest cryptographic techniques and is selected as the base algorithm. Although it is easy to understand and implement, extensive research shows that the traditional Caesar cipher suffers from several serious security weaknesses. The most significant limitation is its very small key space, which makes it highly vulnerable to brute-force attacks. An attacker can easily try all possible shifts and recover the original plaintext.

Another major weakness of the Caesar cipher is its reliance on a single fixed key. This results in a predictable substitution pattern where the same plaintext letter always produces the same ciphertext letter. As a result, the cipher is highly susceptible to frequency analysis attacks. Additionally, the traditional Caesar cipher does not consider the position of characters within the plaintext, which reduces randomness and makes the encryption pattern easier to detect.

Furthermore, the Caesar cipher lacks any form of transposition, meaning the order of characters in the plaintext remains unchanged in the ciphertext. This exposes structural patterns of the original message. The algorithm also operates only at the alphabetic level and does not involve any logical or binary-level operations, which further limits its resistance against modern cryptanalysis techniques.

Due to these limitations, it becomes necessary to modify the traditional Caesar cipher using additional mathematical and logical operations. By introducing position-based logic, multiple keys, transposition techniques, and binary-level logical operations, the overall security of the cipher can be significantly enhanced. These modifications aim to increase confusion and diffusion in the ciphertext while maintaining simplicity and reversibility.

3.2) Background for the Newly Introduced Mathematical and Logical Operations

To address the weaknesses identified in the traditional Caesar cipher, a combination of mathematical and logical operations has been introduced in the proposed cryptographic

algorithm. These operations are selected based on their simplicity, reversibility, and effectiveness in improving security.

From a mathematical operation perspective, the enhanced algorithm still utilizes modular arithmetic like the traditional Caesar cipher. Alphabetic shifting is performed using modulo-26 operations to ensure that character values remain within the valid alphabet range. However, instead of using a single constant shift value, multiple keys are applied to increase complexity. The use of forward and backward shifts ensures that encryption remains reversible while reducing predictability.

For encryption,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

For Decryption,

$P_i = (C_i - K_1 - K_2) \bmod 26$ (if the position of Ciphertext letter is in odd position)

$P_i = (C_i - K_1 + K_2) \bmod 26$ (if the position of Ciphertext letter is in even position)

In addition, a Rail Fence transposition technique is incorporated to rearrange the order of characters after substitution. Transposition adds diffusion by altering character positions, which helps hide structural patterns present in the original message.

Finally, binary-level operation using the XOR gate is applied. Characters are converted into 8-bit ASCII binary format, and a fixed XOR key is used to perform bitwise XOR operations. XOR is chosen due to its simplicity and reversible nature, making it suitable for both encryption and decryption. This binary-level processing introduces an additional layer of security beyond classical substitution and transposition techniques.

Together, these mathematical and logical operations form the foundation of the newly developed cryptographic algorithm, significantly improving its resistance to basic cryptanalysis techniques while preserving computational simplicity.

3.3) Encryption and Decryption from newly made modification

The encryption process of the Cipher is Carried out in three sequential stages. Each stage adds an additional layer of security by introducing mathematical, logical, positional, and bitwise transformations.

Example:

- Plaintext: PRADIP (A=0 to Z=25)
- Main key (K1) = 5 (The main key is added to all characters)
- Logical Key (K2) = 3 (Key 2 is added or subtracted based on the position of the characters in the plaintext.)
 - Odd Position = +3
 - Even Position = -3

Finally, modulo 26 is applied to keep the result within the alphabet range.

Step 1) Encrypting Plaintext: PRADIP

$C_i = (P_i + K1 + K2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K1 - K2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	P	15	$15+5+3$	23	X
2 (even)	R	17	$17+5-3$	19	T
3 (odd)	A	0	$0+5+3$	8	I
4 (even)	D	3	$3+5-3$	5	F
5 (odd)	I	8	$8+5+3$	16	Q
6 (even)	P	15	$15+5-3$	17	R

Table 4: Encryption Process of Dual-key Logical Caesar Cipher with Rail Fence and XOR

Generated Cipher text: XTIFQR

Step 2) Rail Fence Cipher (2 Rails)

The output of the Caesar cipher is rearranged using a 2-rail Rail Fence transposition, which changes the order of characters to improve diffusion.

Rail 1	X		I		Q	
Rail 2		T		F		R

Table 5 : Rail Fence Encryption of Dual-key Logical Caesar cipher with Rail Fence and XOR

Generated Ciphertext (2): XIQTFR

Step 3: ASCII-Binary and XOR Operation

Each Character is converted into its 8-bit ASCII binary form. A fixed XOR KEY (00001111) is applied bit-by-bit. The generated 8bit binary after XOR is changed into decimal than ASCII value is kept according to generated decimal.

XOR Operation Table

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII	Final Cipher text
X	88	01011000	00001111	01010111	87	W
I	73	01001001	00001111	01000110	70	F
Q	81	01010001	00001111	01011110	94	^
T	84	01010100	00001111	01011011	91	[
F	70	01000110	00001111	01001001	73	I
R	82	01010010	00001111	01011101	93	]

Table 6: XOR operation of Dual-Key logical Caesar Cipher with Rail Fence and XOR

Final Ciphertext: WF^[I]

Now, Decrypting the Ciphertext,

Step 1: XOR Decryption

The ciphertext is converted into ASCII binary and XORed again with the same XOR key. Since XOR is symmetric,

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII	Plain text (1)
W	87	01010111	00001111	01011000	88	X
F	70	01000110	00001111	01001001	73	I
^	94	01011110	00001111	01010001	81	Q
[	91	01011011	00001111	01010100	84	T
l	73	01001001	00001111	01000110	70	F
]	93	01011101	00001111	01010010	82	R

Table 7: Decryption process of XOR

WF^[I] → XIQTFR

Step 2: Rail Fence Decryption

Using the same number of rails (2), the zig-zag pattern is reconstructed, and characters are read in the original order.

Rail 1	X		I		Q	
Rail 2		T		F		R

Table 8: Rail Fence Decryption

XIQTFR → XTIFQR

Step 3: Mathematical Caesar Cipher

Each character is converted to its alphabet position and operations are reversed:

- $P_i = (C_i - K_1 - K_2) \bmod 26$ (for odd position)
- $P_i = (C_i - K_1 + K_2) \bmod 26$ (for even position)

Modulo 26 is applied to maintain valid alphabet range

Decryption Table:

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	X	23	$23-5-3$	15	P
2 (even)	T	19	$19-5+3$	17	R
3 (odd)	I	8	$8-5-3$	0	A
4 (even)	F	5	$5-5+3$	3	D
5 (odd)	Q	16	$16-5-3$	8	I
6 (even)	R	17	$17-5+3$	15	P

Final Plaintext: PRADIP

The proposed algorithm enhances the traditional Caesar cipher by introducing position-based logical operations, transposition using Rail Fence, and a binary-level XOR operation, significantly improving confusion and diffusion while maintaining reversibility.

3.5) Dual-Key Logical Caesar Cipher with Rail Fence and XOR

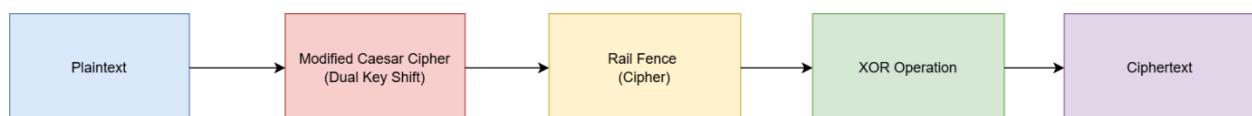


Figure 8: Architecture of Dual-Key Caesar Cipher with Rail Fence and XOR:

3.5.1) Why was the Modification Necessary?

The traditional Caesar Cipher suffers from several major security weaknesses. It uses a single fixed key, has a very small key space, and preserves the original letter frequency and character order, making it highly vulnerable to brute-force and frequency analysis attacks. Additionally, the classical Caesar Cipher does not consider the position of characters in the plaintext and operates only at the alphabetic level.

To overcome these limitations, modifications were necessary to increase complexity, introduce positional dependency, and add both transposition and binary-level logical operations. These improvements aim to enhance confusion and diffusion while keeping the algorithm simple and educational.

3.5.2) The New Methodology Implied

The proposed cryptographic algorithm enhances the Caesar Cipher by combining Mathematical, Logical and transpositional techniques. The methodology follows a multi-layered approach:

1. A dual-key Caesar Cipher is applied, where:
 - The main key (K1) is applied to all Characters.
 - A logical key (K2) is applied based on the position of characters. If the position of the character is odd, then for the encryption part we do (+k2) and if the position of the character is at even then we do (-k2) at the encryption part. So, based on the position of the characters the add and subtraction (Mathematical Operation) part is done.
2. A Rail Fence transposition technique is used to rearrange character position and break the ciphertext which was generated from the above Caesar Cipher. After this technique a new Ciphertext will get generated.
3. The transposed text which was generated from Rail Fence will now get converted into a 8-Bit ASCII binary and after this with a fixed XOR key every ASCII generated binary will be compared with XOR to create a new decimal then according to the generate decimal it will be compared in the ASCII table to see the character of the decimal. This is introduced to have a logical operation which applies binary-level security.

3.5.3) The New Encryption Algorithm

Step 1: Read Plaintext

Step 2: Assign Character Position

Step 3: Applying Modified Caesar Cipher

Step 4: Convert each character in the plaintext to its alphabetical index

Step 5: Apply the main key (K1)

Step 6: Apply the logical key (k2) based on the position

Step 7: If the position is odd, shift forward (+k2)

Step 8: If the position is even, shift backward (-k2)

Step 9: Apply modulo 26 operations to keep values within alphabet range.

Step 10: Convert the result back to a Character.

Step 11: Applying Rail Fence Transposition to the generated Ciphertext

Step 12: Convert each character of the transposed text into its 8-bit ASCII binary representation

Step 13: Applying XOR Logical Operation between each 8-bit binary value and the XOR key (k3)

Step 14: Convert the resulting binary values back to character

Step 15: Final Ciphertext is generated

3.5.4) The New Decryption Algorithm

Step 1: Read Ciphertext

Step 2: Convert each character into its 8-bit binary ASCII binary form.

Step 3: Applying XOR logical operation with the same XOR key (k3) to each binary value

Step 4: Convert the result back to characters

Step 5: Reversing the Rail Fence Transposition

Step 6: Reversing Modified Caesar Cipher for each character.

Step 7: Identifying the position of the Cipher text

Step 8: Reverse the logical key (k2)

Step 9: If the position is odd, shift (backward) (-k2)

Step 10: If position is even, shift (forward) (+k2)

Step 11: Reversing the main key k1 (shift backward)

Step 12: Apply modulo 26 operations to keep the alphabet in range

Step 13: Convert back to plain text characters

Step 14: The original Plaintext (message is generated)

3.5.5) Flow chart of encryption

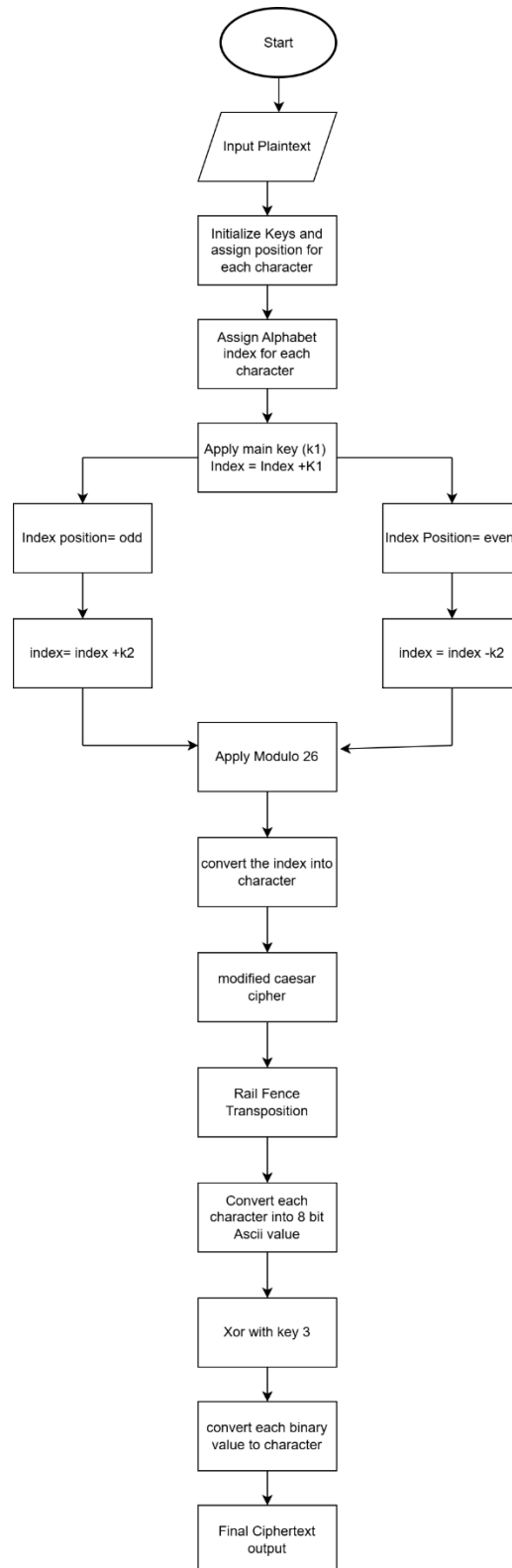


Figure 9 : Encryption Process of Dual-key Logical Caesar Cipher with Rail Fence and XOR

3.5.6) Flowchart of Decryption

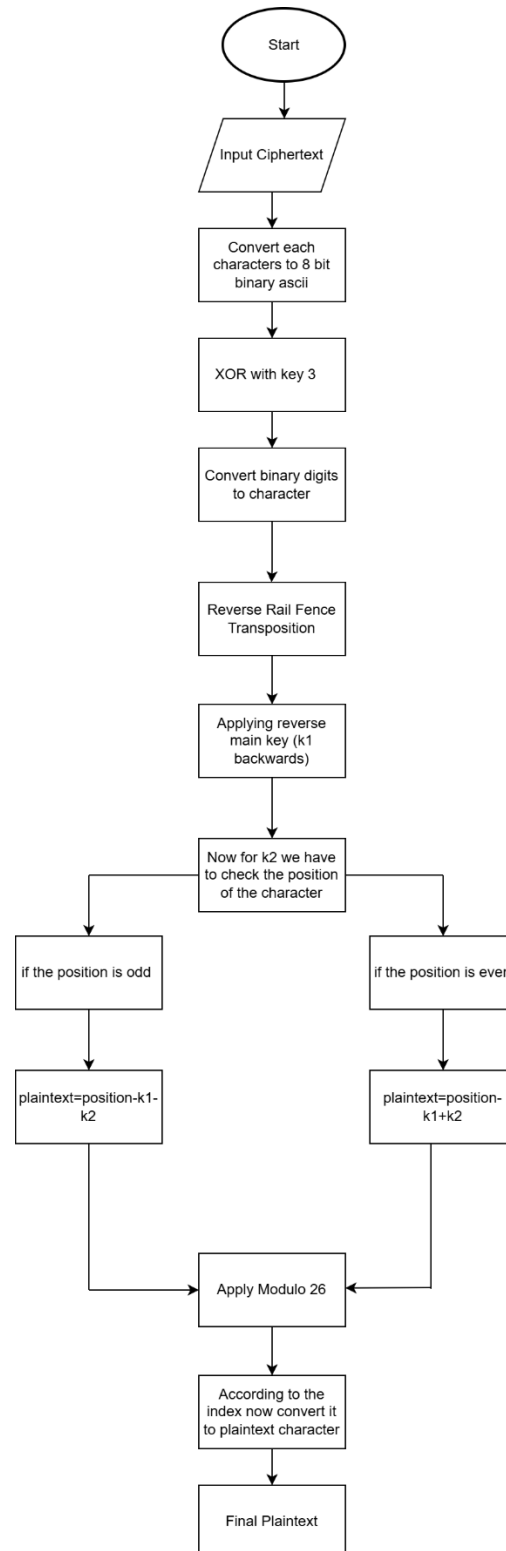


Figure 10: Decryption Process of Dual Key Logical Caesar Cipher with Rail Fence and XOR

4) Testing

4.1) Test 1: For test 1 Plaintext **\$TUDENT** is taken where, fixed key is 4 and logical key is 3 and XOR key is 10101010.

Given,

- Plaintext: Student
- Key (1): 4
- Key (2): 3
- XOR key: 10101010

Encryption Process:

Step 1: Encrypting the given plaintext,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	\$	No position cause not an alphabet	No operation applied		\$
2 (even)	T	19	$19+4-3$	20	U
3 (odd)	U	20	$20+4+3$	1	B
4 (even)	D	3	$3+4-3$	4	E
5 (odd)	E	4	$4+4+3$	11	L
6 (even)	N	13	$13+4-3$	14	O
7(odd)	T	19	$19+4+3$	0	A

Table 9:Modified Caesar Cipher Encryption process of test 1

Generated Ciphertext: \$UBELOA

Step 2: Encrypting the cipher text with Rail Fence (2 Rails)

Rail 1	\$		B		L		A
Rail 2		U		E		O	

Table 10: Rail Fence Encryption of Test 1

Generated Ciphertext: ZBLAUEO

Step 3: ASCII - Binary and XOR Operation

XOR key: 10101010

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Final Ciphertext
\$	36	00100100	10101010	10001110	142	Ž
B	66	01000010	10101010	11101000	232	è
L	11	00001011	10101010	10100001	161	ı
A	0	00000000	10101010	10101010	170	ª
U	85	01010101	10101010	11111111	255	ÿ
E	69	01000101	10101010	11101111	239	ï
O	79	01001111	10101010	11100101	229	å

Table 11: XOR encryption of Test 1

Final Ciphertext: Ž èı ªÿå

Decryption Process:

Step 1: XOR Decryption

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII	Plain text (1)
Ž	142	10001110	10101010	00100100	36	\$
è	232	11101000	10101010	01000010	66	B
ı	161	10100001	10101010	00001011	11	L
ª	170	10101010	10101010	00000000	0	A
ÿ	255	11111111	10101010	01010101	85	U

İ	239	11101111	10101010	01000101	69	E
å	229	11100101	10101010	01001111	79	O

Table 12: XOR decryption of test 1

Generated Plaintext= \$BLAUEO

Step 2: Rail Fence Decryption

Rail 1	\$		B		L		A
Rail 2		U		E		O	

Table 13: Rail Fence Decryption of test 1

Generated Plaintext: ZUBELOA

Step 3: Mathematical Caesar Decryption

- Odd position $\rightarrow -K_2$ then $-K_1$
- Even position $\rightarrow +K_2$ then $-K_1$

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	\$				\$
2 (even)	U	20	$20-4+3$	19	T
3 (odd)	B	1	$1-4-3$	20	U
4 (even)	E	4	$4-4+3$	3	D
5 (odd)	L	11	$11-4-3$	4	E
6 (even)	O	14	$14-4+3$	13	N
7 (odd)	A	0	$0-4-3$	19	T

Table 14: Modified Caesar Decryption of Test 1

Final Plaintext: \$TUDENT

4.2) Test 2: For test 1 Plaintext **TEACHER** is taken where, fixed key is 5 and logical key is 2 and XOR key is 11110000.

Given,

- Plaintext: TEACHER
- Key (1): 5
- Key (2): 2
- XOR key: 11110000

Encryption Process:

Step 1: Encrypting the given plaintext,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	T	19	$19+5+2$	0	A
2 (even)	E	4	$4+5-2$	7	H
3 (odd)	A	0	$0+5+2$	7	H
4 (even)	C	2	$2+5-2$	5	F
5 (odd)	H	7	$7+5+2$	14	O
6 (even)	E	4	$4+5-2$	7	H
7(odd)	R	17	$17+5+2$	24	Y

Table 15: Modified Caesar Cipher Encryption of test 2

Generated Ciphertext: AHHFOHY

Step 2: Encrypting the cipher text with Rail Fence (2 Rails)

Rail 1	A		H		O		Y
--------	---	--	---	--	---	--	---

Rail 2		H		F		H	
--------	--	---	--	---	--	---	--

Table 16: Rail Fence Encryption of Test 2

Generated Ciphertext: AHOYHFH

Step 3: ASCII - Binary and XOR Operation

XOR Key: 11110000

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Final Ciphertext
A	65	01000001	11110000	10110001	177	±
H	72	01001000	11110000	10111000	184	،
O	79	01001111	11110000	10111111	191	¿
Y	89	01011001	11110000	10101001	169	©
H	72	01001000	11110000	10111000	184	،
F	70	01000110	11110000	10110110	182	¶
H	72	01001000	11110000	10111000	184	،

Table 17: XOR Operation encryption of test 2

Generated Final Ciphertext: ±,¿©,¶,،

Decryption Process:

Step 1: XOR Decryption

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Plaintext
±	177	10110001	11110000	01000001	65	A
،	184	10111000	11110000	01001000	72	H
¿	191	10111111	11110000	01001111	79	O
©	169	10101001	11110000	01011001	89	Y
،	184	10111000	11110000	01001000	72	H
¶	182	10110110	11110000	01000110	70	F
،	184	10111000	11110000	01001000	72	H

Table 18: XOR Decryption of test 2

Generated Plaintext: AHOYHFH

Step 2: Rail Fence Decryption

Rail 1	A		H		O		Y
Rail 2		H		F		H	

Table 19: Rail Fence Decryption of test 2

Generated Plaintext: AHHFOHY

Step 3: Mathematical Caesar Decryption

- $P_i = (C_i - K_1 - K_2) \bmod 26$ (for odd position)
- $P_i = (C_i - K_1 + K_2) \bmod 26$ (for even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	A	0	0-5-2	19	T
2 (even)	H	7	7-5+2	4	E
3 (odd)	H	7	7-5-2	0	A
4 (even)	F	5	5-5+2	2	C
5 (odd)	O	14	14-5-2	7	H
6 (even)	H	7	7-5+2	4	E
7 (odd)	Y	24	24-5-2	17	R

Table 20: Modified Caesar Decryption of test 2

Final Plaintext: TEACHER

4.3) Test 3: For test 3 Plaintext **COMPUTING** is taken where, fixed key is 3 and logical key is 4 and XOR key is

Given,

- Plaintext: COMPUTING
- Key (1): 3
- Key (2): 4

- XOR key: 00001111

Encryption Process:

Step 1: Encrypting the given plaintext,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	C	2	2+3+4	9	J
2 (even)	O	14	14+3-4	13	N
3 (odd)	M	12	12+3+4	19	T
4 (even)	P	15	15+3-4	14	O
5 (odd)	U	20	20+3+4	1	B
6 (even)	T	19	19+3-4	18	S
7(odd)	I	8	8+3+4	15	P
8 (even)	N	13	13+3-4	12	M
9 (odd)	G	6	6+3+4	13	N

Generated Ciphertext: JNTOBSPMN

Table 21: Modified Caesar encryption test 3

Step 2: Encrypting the cipher text with Rail Fence (2 Rails)

Rail 1	J		T		B		P		N
Rail 2		N		O		S		M	

Table 22 : Rail Fence Encryption of test 3

Generated Ciphertext: JTBPNNOSM

Step 3: ASCII - Binary and XOR Operation

XOR Key: 00001111

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Final Ciphertext
J	74	01001010	00001111	01000101	69	E
T	84	01010100	00001111	01011011	91	[
B	66	01000010	00001111	01001101	77	M
P	15	01010000	00001111	01011111	95	_
N	66	01001110	00001111	01000001	65	A
N	78	01001110	00001111	01000001	65	A
O	79	01001111	00001111	01000000	64	@
S	83	01010011	00001111	01011100	92	\
M	77	01001101	00001111	01000010	66	B

Table 23 : XOR Operation encryption of test 3

Generated Final Ciphertext: E[M_AA@]B

Decryption Process:

Step 1: XOR Decryption

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Plaintext
E	69	01000101	00001111	01001010	74	J
[	91	01011011	00001111	01010100	84	T
M	77	01001101	00001111	01000010	66	B
_	95	01011111	00001111	01010000	15	P
A	65	01000001	00001111	01000010	66	N
A	65	01000001	00001111	01001110	78	N
@	64	01000000	00001111	01001111	79	O
\	92	01011100	00001111	01010011	83	S
B	66	01000010	00001111	01001101	77	M

Table 24: XOR Operation decryption of test 3

Generated Plaintext: JTBPNNOSM

Step 2: Rail Fence Decryption

Rail 1	J		T		B		P		N
Rail 2		N		O		S		M	

Table 25: Rail Fence decryption of test 3

Generated Plaintext: JNTOBSPMN

Step 3: Mathematical Caesar Decryption

- $P_i = (C_i - K_1 - K_2) \bmod 26$ (for odd position)
- $P_i = (C_i - K_1 + K_2) \bmod 26$ (for even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	J	9	9-3-4	2	C
2 (even)	N	13	13-3+4	14	O
3 (odd)	T	19	19-3-4	12	M
4 (even)	O	14	14-3+4	15	P
5 (odd)	B	1	1-3-4	20	U
6 (even)	S	18	18-3+4	19	T
7 (odd)	P	15	15-3-4	8	I
8 (even)	M	12	12-3+4	13	N
9 (odd)	N	13	13-3-4	6	G

Table 26: Modified Caesar Decryption of test 3

Final Plaintext: COMPUTING

4.4) Test 4: For test 4 Plaintext **BUTTERFLY** is taken where, fixed key is 6 and logical key is 1 and XOR key is 01010101

Given,

- Plaintext: BUTTERFLY
- Key (1): 6

- Key (2): 1
- XOR key: 01010101

Encryption Process:

Step 1: Encrypting the given plaintext,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	B	1	$1+6+1$	8	I
2 (even)	U	20	$20+6-1$	25	Z
3 (odd)	T	19	$19+6+1$	0	A
4 (even)	T	19	$19+6-1$	24	Y
5 (odd)	E	4	$4+6+1$	11	L
6 (even)	R	17	$17+6-1$	22	W
7(odd)	F	5	$5+6+1$	12	M
8 (even)	L	11	$11+6-1$	16	Q
9 (odd)	Y	24	$24+6+1$	5	F

Table 27: Modified Caesar Encryption of test 4

Generated Ciphertext: IZAYLWMQF

Step 2: Encrypting the cipher text with Rail Fence (2 Rails)

Rail 1	I		A		L		M		F
Rail 2		Z		Y		W		Q	

Table 28: Rail Fence Encryption

Generated Ciphertext: IALMFZYWQ

Step 3: ASCII - Binary and XOR Operation

XOR Key: 00001111

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Final Ciphertext
I	73	01001001	00001111	01000110	70	F
A	65	01000001	00001111	01001110	78	N
L	76	01001100	00001111	01000011	67	C
M	77	01001101	00001111	01000010	66	B
F	70	01000110	00001111	01001001	73	I
Z	90	01011010	00001111	01010101	85	U
Y	89	01011001	00001111	01010110	86	V
W	87	01010111	00001111	01011000	88	X
Q	81	01010001	00001111	01011110	94	^

Table 29: XOR operation encryption of test 4

Generated Final Ciphertext: FNCBIUVX^

Decryption Process:

Step 1: XOR Decryption

Character	ASCII	8-bit binary	XOR KEY	XOR Result	Decimal	Plaintext
F	70	01000110	00001111	01001001	73	I
N	78	01001110	00001111	01000001	65	A
C	67	01000011	00001111	01001100	76	L
B	66	01000010	00001111	01001101	77	M
I	73	01001001	00001111	01000110	70	F
U	85	01010101	00001111	01011010	90	Z
V	86	01010110	00001111	01011001	89	Y
X	88	01011000	00001111	01010111	87	W

^	94	01011110	00001111	01010001	81	Q
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Table 30: XOR Operation decryption process of test 4

Generated Plaintext: IALMFZYWQ

Step 2: Rail Fence Decryption

Rail 1	I		A		L		M		F
Rail 2		Z		Y		W		Q	

Table 31: Rail Fence decryption of test 4

Generated Plaintext: IZAYLWMQF

Step 3: Mathematical Caesar Decryption

- $P_i = (C_i - K_1 - K_2) \bmod 26$ (for odd position)
- $P_i = (C_i - K_1 + K_2) \bmod 26$ (for even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	I	8	$8-6-1$	1	B
2 (even)	Z	25	$25-6+1$	20	U
3 (odd)	A	0	$0-6-1$	19	T
4 (even)	Y	24	$24-6+1$	19	T
5 (odd)	L	11	$11-6+1$	4	E
6 (even)	W	22	$22-6+1$	17	R
7 (odd)	M	12	$12-6-1$	5	F
8 (even)	Q	16	$16-6+1$	11	L
9 (odd)	F	5	$5-6+1$	24	Y

Table 32: Modified Caesar Decryption of test 4

Final Plaintext: BUTTERFLY

4.5) TEST 5: For test 5 Plaintext **COMFORTABLE** is taken where, fixed key is 2 and logical key is 5 and XOR key is 00110011

Given,

- Plaintext: COMFORTABLE
- Key (1): 2
- Key (2): 5
- XOR key: 00110011

Encryption Process:

Step 1: Encrypting the given plaintext,

$C_i = (P_i + K_1 + K_2) \bmod 26$ (if the position of plaintext letter is in odd position)

$C_i = (P_i + K_1 - K_2) \bmod 26$ (if the position of plaintext letter is in even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Cipher text (1)
1 (odd)	C	2	2+2+5	9	J
2 (even)	O	14	14+2-5	11	L
3 (odd)	M	12	12+2+5	19	T
4 (even)	F	5	5+2-5	2	C
5 (odd)	O	14	14+2+5	21	V
6 (even)	R	17	17+2-5	14	O
7(odd)	T	19	19+2+5	0	A
8(even)	A	0	0+2-5	23	X
9(odd)	B	1	1+2+5	8	I
10(even)	L	11	11+2-5	8	I
11(odd)	E	4	4+2+5	11	L

Table 33: Modified Caesar Encryption of test 5

Generated Ciphertext: JLTCVOAXIIL

Step 2: Encrypting the cipher text with Rail Fence (2 Rails)

Rail 1	J		T		V		A		I		L
Rail 2		L		C		O		X		I	

Table 34: Rail Fence encryption of test 5

Generated Ciphertext: JTVAILLCOXI

Step 3: ASCII - Binary and XOR Operation

XOR Key:00110011

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Final Ciphertext
J	74	01001010	00110011	01111001	121	y
T	84	01010100	00110011	01100111	103	g
V	86	01010110	00110011	01100101	101	e
A	65	01000001	00110011	01110010	114	r
I	73	01001001	00110011	01111010	122	z
L	76	01001100	00110011	01111111	127	DEL
L	76	01001100	00110011	01111111	127	DEL
C	67	01000011	00110011	01110000	112	p
O	79	01001111	00110011	01111100	124	l
X	88	01011000	00110011	01101011	107	k
I	73	01001001	00110011	01111010	122	z

Table 35: XOR operation encryption of test 5

Generated Final Ciphertext: ygerzDELDELplkz

Decryption Process:

Step 1: XOR Decryption

Character	ASCII	8-bit binary	XOR KEY	XOR Result	ASCII Decimal	Plaintext
y	121	01111001	00110011	01001010	74	J
g	103	01100111	00110011	01010100	84	T
e	101	01100101	00110011	01010110	86	V
r	114	01110010	00110011	01000001	65	A
z	122	01111010	00110011	01001001	73	I
DEL	127	01111111	00110011	01001100	76	L
DEL	127	01111111	00110011	01001100	76	L
p	112	01111100	00110011	01000011	67	C
l	124	01111100	00110011	01001111	79	O
k	107	01101011	00110011	01011000	88	X
z	122	01111010	00110011	01001001	73	I

Table 36: Decryption process XOR

Generated Plaintext: JTVAILLCOXI

Step 2: Rail Fence Decryption

Rail 1	J		T		V		A		I		L
Rail 2		L		C		O		X		I	

Table 37: Rail Fence decryption of test 5

Generated Plaintext: JTVAILLCOXI

Step 3: Mathematical Caesar Decryption

- $P_i = (C_i - K_1 - K_2) \bmod 26$ (for odd position)
- $P_i = (C_i - K_1 + K_2) \bmod 26$ (for even position)

Position of Plaintext	Character of Plaintext	Alphabet Position	Operation Applied	Final Position	Final Plaintext
1 (odd)	J	9	9-2-5	2	C
2 (even)	L	11	11-2+5	14	O
3 (odd)	T	19	19-2-5	12	M
4 (even)	C	2	2-2+5	5	F
5 (odd)	V	21	21-2-5	14	O
6 (even)	O	14	14-2+5	17	R
7 (odd)	A	0	0-2-5	19	T
8(even)	X	23	23-2+5	0	A
9(odd)	I	8	8-2-5	1	B
10(even)	I	8	8-2+5	11	L
11(odd)	L	11	11-2-5	4	E

Table 38: Modified Caesar Decryption of test 5

Final Plaintext: COMFORTABLE

5)Evaluation

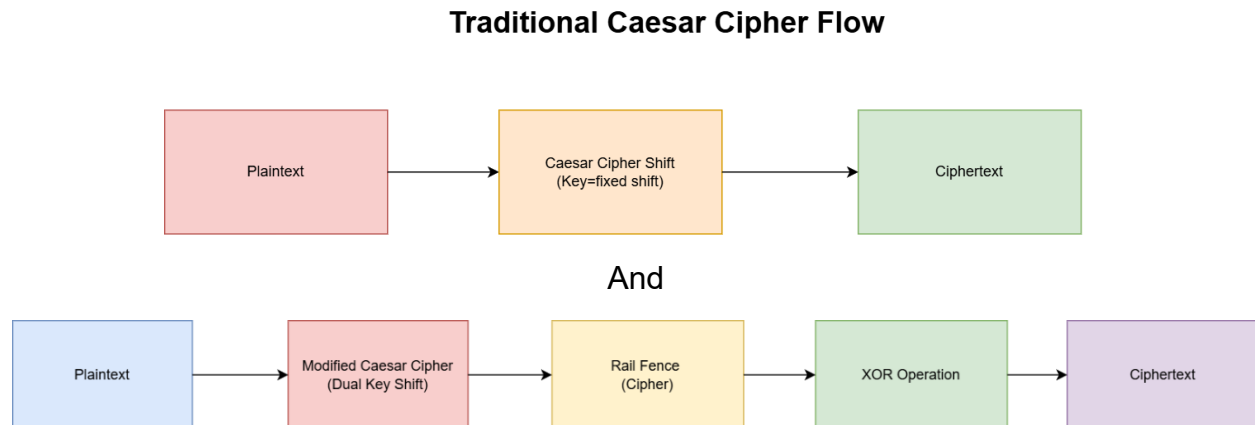


Figure 11 : Comparison between traditional cipher and modified cipher

The traditional Caesar cipher is a simple substitution technique where each plaintext letter is shifted by a fixed number of positions in the alphabet. Although easy to implement, it provides very weak security due to predictable patterns and a limited key space.

But the modified cryptographic algorithm improves the traditional Caesar cipher by applying multiple layers of transformation. It uses position-based key operations, Rail Fence transposition, and a binary XOR process to increase complexity and reduce predictability while still allowing correct decryption.

5.1) Strengths of the Modified Cipher

- Uses multiple encryption layers instead of a single shift operation.
- It applies different operations for odd and even character positions.
- Reduces visible plaintext patterns in the ciphertext.
- Improves confusion and diffusion through transposition and XOR.
- Ensures reversible and accurate decryption.
- Simple to implement while still being more secure than basic ciphers.

5.2) Weakness of the Modified Cipher

- Limited key size makes brute-force attacks possible
- Not resistant to modern cryptographic attack techniques
- Fixed Rail Fence structure reduces transposition strength

- No secure key generation or key management mechanism
- Not suitable for encrypting sensitive or large-scale data
- XOR key remains constant, making it predictable over time

Overall, the modified cryptographic algorithm successfully strengthens the traditional Caesar Cipher by introducing logical, positional, and binary-level transformations. But it still doesn't meet modern security standards.

6) Conclusion

This coursework was carried out to understand the concept of cryptography and to design and test a new cryptographic algorithm based on a classical cipher. Through detailed research, the fundamentals of cryptography, encryption, decryption, and the CIA triad were studied, which helped in building a strong theoretical background for this coursework.

The Caesar Cipher was selected as the base algorithm due to its simplicity and historical importance. However, during the study, it was clearly observed that the traditional Caesar Cipher has many security weaknesses, such as a small key space, fixed shifting pattern, and vulnerability to brute-force and frequency analysis attacks. Because of these limitations, it was necessary to modify the algorithm to improve its security.

To overcome these weaknesses, a new modified cryptographic algorithm was developed by introducing position-based logical operations, dual keys, Rail Fence transposition, and a binary-level XOR operation. These additional steps increased the complexity of the encryption process and helped in reducing predictable patterns in the ciphertext. The encryption and decryption process was carried out step by step, and the results showed that the original plaintext could be successfully recovered using the correct keys and reverse operations.

Testing using multiple examples confirmed that the algorithm works correctly and produces different ciphertext outputs for different plaintext inputs. The evaluation also showed that the modified cipher is more secure than the traditional Caesar Cipher, especially in terms of confusion and diffusion.

However, the algorithm still has some limitations. The key size is limited, the Rail Fence structure is fixed, and the XOR key remains constant. Therefore, the algorithm does not fully meet modern cryptographic security standards and is not suitable for protecting highly sensitive information.

In conclusion, this coursework successfully demonstrates how a simple classical cipher can be enhanced using mathematical, logical, and binary-level operations. Although the proposed algorithm is mainly suitable for educational and learning purposes, it provides a clear understanding of cryptographic design, implementation, testing, and evaluation.

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